

# EEEP 326.

## Power Courses Electrical Power Systems III

| PROGRAMME: HND IN ELECTRICAL/ELECTRONIC ENGINEERING TECHNOLOGY (POWER AND MACHINES OPTION) |                                                                                                      |                                                                                                          |           |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------|
| Course: ELECTRIC POWER SYSTEMS III                                                         | Course Code: <del>EEEP326</del>                                                                      | Contact Hours: 75 HOURS                                                                                  |           |
| Course Specification: Theoretical Content                                                  |                                                                                                      |                                                                                                          |           |
|                                                                                            | General Objective 1.0: Understand inter-connected systems.                                           |                                                                                                          |           |
| WEEK                                                                                       | Specific Learning Outcome:                                                                           | Teachers Activities                                                                                      | Resources |
|                                                                                            | On completion of this course the student should be able to:- Interconnected systems                  | - Use demonstrative problems to assemble admittance matrix for a sample matrix.                          |           |
| ✓                                                                                          | 1.1 Explain interconnected power systems.                                                            | - Use network analysis to demonstrate solution of Load flow problems.                                    |           |
| ✓                                                                                          | 1.2 State the advantages and disadvantages of interconnected power systems.                          | - Encourage students to write simple programmes to set up an admittance matrix of a simple power system. |           |
| ✓                                                                                          | 1.3 Explain the construction of power circle diagrams.                                               | - Demonstrate short circuit fault using a simple LC circuit.                                             |           |
| ✓                                                                                          | 1.4 Explain the techniques for reducing interconnected systems to simple equivalents.                | - Use problems to illustrate symmetrical and unsymmetrical faults.                                       |           |
| ✓                                                                                          | 1.5 Solve problems involving 1.3 and 1.4 above.                                                      | - Use demonstrative problems to illustrate disruptive critical voltage for an overhead line.             |           |
| ✓                                                                                          | 1.6 Formulate the nodal admittance matrices of various networks. (236)                               | - Use problems to show the application of the surge velocity equation in a uniform line.                 |           |
| 1-5                                                                                        | 1.7 Explain the need for load flow studies. (225)                                                    |                                                                                                          | 182 book  |
|                                                                                            | 1.8 State load-flow problem.                                                                         |                                                                                                          |           |
|                                                                                            | 1.9 Outline the variable (P, Q, V/S affecting load flow in a power system network. (197)             |                                                                                                          |           |
|                                                                                            | 1.10 Classify the variables in 1.9 into control independent and dependent variables.                 |                                                                                                          |           |
|                                                                                            | 1.11 Derive the general form of the load-flow equation in:<br>i. Rectangular form;<br>ii Polar form; |                                                                                                          |           |
|                                                                                            | 1.12 Know one method of load flow solution [ ]                                                       |                                                                                                          |           |
|                                                                                            | 1.13 Calculate load flow analysis of interconnected systems.                                         |                                                                                                          |           |
|                                                                                            | 1.14 Explain the application of digital computers to load flow studies.                              |                                                                                                          |           |

9-11  
Stability book

30 --4  
51 --5  
61 --5  
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PROGRAMME: HND IN ELECTRICAL/ELECTRONIC ENGINEERING TECHNOLOGY (POWER AND MACHINES OPTION)

Course: ELECTRIC POWER SYSTEMS III

Course Code: EEE 326

Contact Hours:  
75 HOURS

Course Specification: Theoretical Content

|      | 1.15 Perform load flow analysis of interconnected systems,<br>1.16 Write a computer program to perform load flow analysis of a simple power network.<br>1.17 Execute item 1.6 in a computer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                               |           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------|
|      | General Objective 2.0: Understand the performance of fault analysis of interconnected systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                               |           |
| WEEK | Specific Learning Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Teachers Activities                           | Resources |
|      | 2.1 Explain the various types of faults that occur on generators and transformers.<br>2.2 State the various types of faults that occur on transmission lines.<br>2.3 Explain short circuit, open circuit and earth faults on lines.<br><u>2.4 Define transient and subtransient reactances.</u><br>2.5 Explain subtransient and transient reactances using the appropriate wave form of a faulted generator.<br>2.6 Draw typical wave forms of short circuit currents in power systems.<br>2.7 Define the peak short circuit current (dynamic) using the wave form of 2.6 above.<br>2.8 Explain a symmetrical fault.<br>2.9 Solve symmetrical fault problems using line diagrams and per unit method. | - Outline the remedies for the various faults |           |

analysis of  
1.15-1.17  
→

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CE423

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$$V = \frac{E_a}{I}$$

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75 HOURS

**Course Specification: Theoretical Content**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                             |                                                                                                                                             |                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <p>2.10 Explain positive, negative and zero sequence components.</p> <p>2.11 Derive expressions for the symmetrical components mentioned in 2.10 in terms of the line values.</p> <p>2.12 State the expression for power in symmetrical components.</p> <p>2.13 Define unsymmetrical faults (single line to ground, double line to ground, etc.)</p> <p>2.14 Explain unsymmetrical fault problems using the symmetrical component networks.</p> <p>2.15 Identify the sequence impedances of power system components. <math>42 \text{ pfs}</math></p> <p>2.16 Find the sequence network for a given power system.</p> <p>2.17 Solve unsymmetrical fault problems using the principles of symmetrical components.</p> <p>2.18 Calculate the MVA fault level on typical power systems.</p> <p>2.19 Explain methods of selecting switchgear, bus-bars, fuses for typical fault levels.</p> | <p>2.10 ✓</p> <p>2.11 ✓</p> <p>2.12 ✓</p> <p>2.13 ✓</p> <p>2.14 ✓</p> <p>2.15 ✓</p> <p>2.16 ✓</p> <p>2.17 ✓</p> <p>2.18 ✓</p> <p>2.19 ✓</p> | <p>2.10 ✓</p> <p>2.11 ✓</p> <p>2.12 ✓</p> <p>2.13 ✓</p> <p>2.14 ✓</p> <p>2.15 ✓</p> <p>2.16 ✓</p> <p>2.17 ✓</p> <p>2.18 ✓</p> <p>2.19 ✓</p> | <p>2.10 ✓</p> <p>2.11 ✓</p> <p>2.12 ✓</p> <p>2.13 ✓</p> <p>2.14 ✓</p> <p>2.15 ✓</p> <p>2.16 ✓</p> <p>2.17 ✓</p> <p>2.18 ✓</p> <p>2.19 ✓</p> |
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PROGRAMME: HND  
MACHINES OPTION)

Course: ELECTRIC POWER SYSTEMS III

Course Code: EEE  
326

75 HOURS

**Course Specification: Theoretical-Content**

General Objective 3.0: Know the system Overvoltages and insulation requirements

| WEEK                            | Specific Learning Outcome:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Teachers Activities                            | Resources |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------|
| <i>Note</i><br>Wk 2<br>1st time | 3.1 Explain corona and factors affecting it. <b>1st time</b><br>3.2 State the various effects of corona<br>3.3 Derive a formula for the disruptive critical voltage for an overhead line.<br>3.4 Explain the causes of over-voltages in power systems.<br>3.5 Draw a typical surge wave form.<br>3.6 List the possible effects of a travelling wave on a transmission system.<br>3.7 Derive an equation for the surge velocity in a uniform line.<br>3.8 Solve, using 3.7, problems on:-<br>a. Surge velocity in a single-phase overhead line in air.<br>b. Surge velocity in a three-phase overhead line in air.<br>c. Surge velocity in a single-phase concentric cable                                    | State practical applications of corona effects |           |
|                                 | 3.9 Deduce an expression for the surge impedance ( $Z_0$ ).<br>3.10 Explain the different types of terminations.<br>3.11 Deduce an expression for reflected surge voltage and current, and transmitted surge voltage and current.<br>3.12 Define the following:<br>a. Reflecting factor of coefficient (p)<br>b. Transmission factor of coefficient (t)<br>3.13 Solve problems on surges using 3.9 and 3.11 above.<br>3.14 Describe the protection of transmission lines against lightning surges.<br>3.15 Draw graph for voltage and current surges when R is greater than $Z_0$ and Explain the graph.<br>3.16 Draw graphs for voltage and current surges when R is less than $Z_0$ and explain the graph. |                                                |           |

**PROGRAMME: HND IN ELECTRICAL/ELECTRONIC ENGINEERING TECHNOLOGY (POWER AND MACHINES OPTION)**

**Course: ELECTRIC POWER SYSTEMS III**

|                         |                       |
|-------------------------|-----------------------|
| <b>Course Code: EEE</b> | <b>Contact Hours:</b> |
| 326                     | 75 HOURS              |

**Course Specification: Theoretical Content**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <p>3.17 Explain the effect of a surge on an overhead line terminated by a transformer.</p> <p>3.18 Deduce expressions for reflected voltage and current surges in an open circuited line and short-circuited line.</p> <p>3.19 Solve problems involving line/cable junctions using the derived equations in 3.18 above.</p> <p>3.20 Describe various types of overhead line insulators and state their uses. <i>reference notes 14/24</i> <b>15/165 (power system)</b></p> <p>3.21 Describe various methods of testing insulators.</p> <p>3.22 Deduce an expression for the voltage distribution across an insulated string. <b>pp 168</b></p> <p>3.23 Solve problems on voltage distribution and string efficiency of insulators. <b>pp 171</b></p> <p>3.24 Explain insulation co-ordination in overhead lines. <b>396</b></p> <p>3.25 Explain the operation of an impulse generator. <b>(paper 3, pdf)</b></p> <p>3.26 Explain the wave shape obtained from an impulse generator</p> |  |  |
| <b>ASSESSMENT:</b> Course work 10%; Course tests 10%; Practical 20%; Examination 60%.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |

**EXPERIMENTS ON ELECTRIC POWER SYSTEM III**

1. Experiments on characteristics of impulse generator
2. Test insulators using various methods.

## Interconnected Power System

Interconnected power system is an electrical grid network for delivering electricity from suppliers to consumers. It consists of "power stations" generating electrical power at lower voltages, "high voltage transmission lines" that carry power from the distant power stations to demand centers at medium voltage level, and finally to "low voltage distribution lines" that connect individual customers.

Power stations may be located near a fuel source, at a dam site, take advantage of renewable energy sources, and are often located away from heavily populated areas.

The electric power which is generated <sup>at a low voltage</sup> is stepped up to a higher voltage (say, 132kV) at which it connects to the electric power transmission network. The bulk power transmission network will move the power long distances, sometimes across international boundaries, until it reaches its wholesale customer (usually the company that owns the local electric power distribution network), say, Yola Electricity Distribution Company (YEDC).

On arrival at a substation, the power will be stepped down from a transmission level voltage to <sup>secondary</sup> voltage distribution level. As it exits the substation, it enters the distribution wiring. Finally, upon arrival at the service location, the power is further stepped down from <sup>secondary</sup> voltage distribution level (say, 33kV) to the required service voltage of 415 V to final consumers. The diagram of Fig. 1.0 gives a schematic diagram of interconnected power system.

①

Schematic Diagram of Interconnected Power System

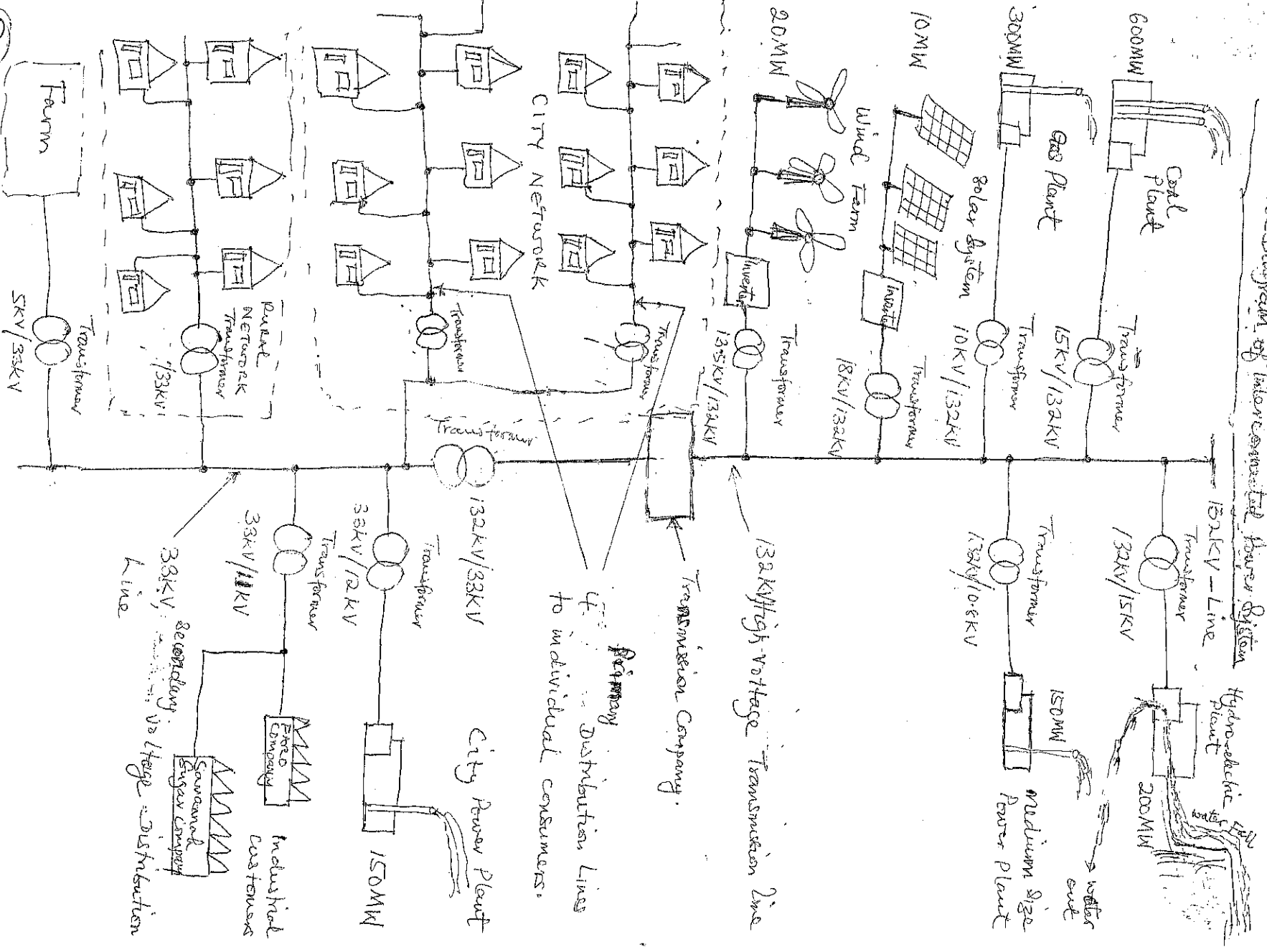
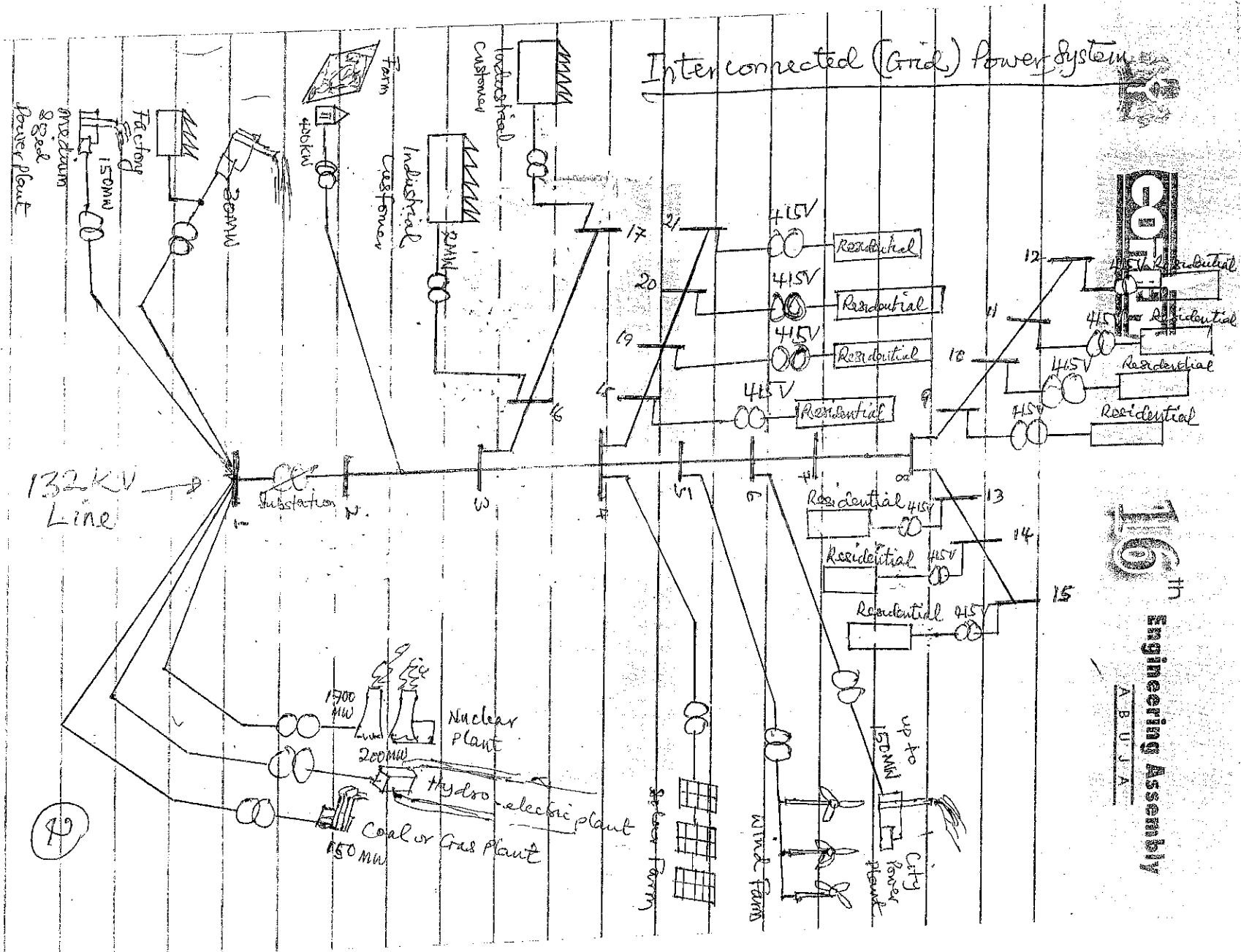


Fig. 1.10: Schematic Diagram of Interconnected Power System



# Interconnected (Grid) power system

16<sup>th</sup>

Engineering Assembly

A B U J A



## Advantages of Interconnected power systems

1. Pooling Reserves: the amount of reserve capacity that must be built by individual networks to ensure reliable operation when demand increases can be reduced by sharing reserves within an interconnected system.
2. Reduced Investment in Generation: individual systems can reduce their generating capacity investment, or postpone the need to add new capacity, if they are able to share generating resources of interconnected system.
3. Improving Load Factor: systems operate most economically when the level of power demand is steady over time, as opposed to having high peaks.

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(3)

~~3~~

4. Diversity of Generation Mix and Supply Security: interconnection between systems that use different technologies and/or fuels to generate electricity provide greater security in the event that one kind of generation becomes limited.

5. Economic Exchange: interconnection allows the dispatch of the least costly generating units within the interconnected area, providing an overall cost savings that can be divided among the component systems. That is, it allows inexpensive power from one system to be sold to systems with more expensive power.

6. Environment and New Plant Siting: interconnections can allow generating units with lower environmental impacts to be used more, and units with higher impacts to be used less. In areas where environmental and land use constraints limit plant construction in less sensitive areas.

7. Coordination of Maintenance Schedules: interconnections permit planned outages of generating and transmission facilities for maintenance to be coordinated so that overall cost and reliability for the interconnected system is optimized.

### 1.3 Disadvantages of Interconnected Power Systems

1. The main disadvantage of the interconnected system is increasing the short circuit <sup>current</sup> level of the system so that occurs at any point in the system, it may spread throughout the whole system, which may lead to a system collapse.
2. Interconnections entail the expense of constructing and operating transmission lines and substations, or in the case of High Voltage Direct Current (HVDC) system, construction of converter stations is involved.

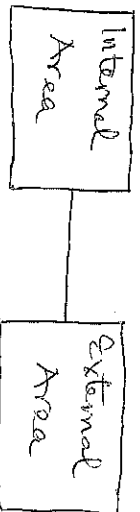
(7)

## Techniques of Reducing Interconnected Systems to Simple Equivalent

Power systems today are colossal networks of interconnected power apparatus often spanning large geographical areas. With the growing trend in facilitating additional interconnections to neighboring systems, the size and complexity of these networks will continue to increase, and thus, bring challenges for planning, operation and control of these networks. Large-scale power system is computationally challenging. Large systems are difficult to handle if not reduced. By using small, equivalent networks, the computational requirements can be significantly reduced. Reduction techniques are useful means of analyzing large, interconnected power systems. The most desirable property of a reduced network is that it should represent the original network as accurately as possible.

Network reduction techniques can be broadly classified as "static" and "dynamic", based on the representation of the model and its intended use. Static reduction analysis are used for power flow calculations during system planning studies. Whereas dynamic technique is used for analysis of system dynamics by using the reduced equivalents of the external areas.

In performing reduction process the power system we want to study is divided into two parts, one part with system variables of interest to us called the study area (or internal area), and another part called the external area, that is only of interest in terms of its effect on the system variables in the internal area as shown in the figure below.



The aim is to reduce the external area to a linear, time-invariant system of lower, while retaining the nonlinear description of the internal area.

## Static Reduction Techniques

Commonly used static reduction techniques are:

1. Ward Reduction,
2. Kron Reduction,
3. Dims Reduction,
4. Zhukov's Reduction, and
5. Modal Reduction.

(a)

### Ward Reduction

The construction of this equivalent starts from the solved model of the entire interconnected system. The injected current  $i(i)$  at each bus  $i$  is obtained from the bus's known complex power  $S(i)$  and voltage  $V(i)$  as follows;

$$i(i) = \frac{S^*(i)}{V^*(i)}$$

Reduction of the network and current vector are performed by using the Gaussian elimination technique.

(b)

### Kron Reduction

This technique, which is actually a form of Gaussian elimination, known as Kron reduction. In this method, instead of arithmetically set the appropriate rows below the diagonal to zero, simply set them to zero below and to the right of a diagonal element by the diagonal element, divide only the elements in the diagonal row by the diagonal element, make ~~the~~ the diagonal element unity, and the same effect obtained ~~obtained~~ in Gaussian elimination will be achieved.

(c)

### Dims Reduction

This reduction method is achieved by the following steps:

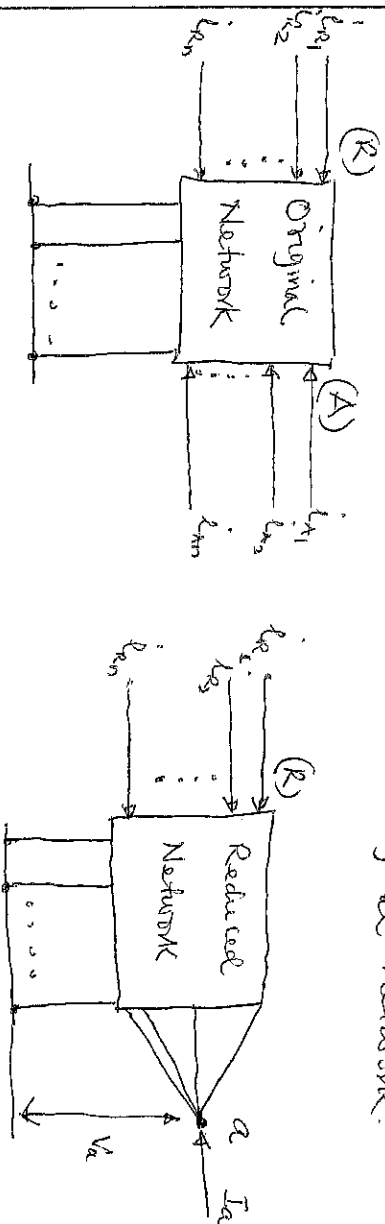
1. Define the set of nodes to be retained and be aggregated.
2. Analyze the modal injections at the non-essential buses and define the aggregating pattern for generators and loads.
3. Assemble zero power balance network (ZPEN).
4. Assemble bus admittance matrix for extended model.
5. Perform Gaussian elimination of non-essential buses.

(11)

(st)

## Zhukov's Reduction

Zhukov's method consists of aggregating the selected nodes through ideal transformers. A set of nodes (A) is aggregated to a single equivalent node (a) as shown in the diagram below. While keeping the node in (R) which retains the current and voltage values and the power injected into node (a) is retained ~~the~~ same in the aggregated node as the original network.



(Zhukov's Reduction Technique.)

(3)

## Modal Reduction

In this method a linear power network is represented with a complete set of differential equations. These equations are reduced to another set of differential equations. This action eliminates specific modes in the linearized models, having little to do with the disturbance. This method is so simple but does not retain the physical structures like generating units, lines, loads etc. It is mostly used for the large scale power systems for transient stability analysis with small disturbances.

## Dynamic Reduction Technique

1.8.2

Dynamic reduction technique commonly used are;

1. The estimate equivalent technique,
2. The coherency equivalent technique.

## (a) Estimate Equivalent Technique

This technique is used in large scaled power system security assessment. The estimate equivalent technique does not need (1.8.2)

(4)

the detail information about the entire structure and parameters of the external system. The procedure for the estimate equivalent is:

1. Perform a disturbance on the system
2. record the system response on the boundary between internal and external areas
3. estimate the parameters of the external system based on the recorded response from the disturbance. In this step several optimal parameter identification methods are used.

### Coherency Equivalent Technique

The coherency equivalent technique is based on identifying the coherent generators by comparison of the generator's rotor angle deviation when a disturbance is applied on to the power network. If the generator rotor angle deviation curves have high similarity, the generators can be aggregated as one by modified generator parameter theoretically. This means the reduction degree mainly depends on the number of coherent generators. When the loads will be aggregated in a dynamic way, the busbars and consequently, the full order power system which contains many busbars can be represented as a reduced system with less generators and busbar number.

### Nodal Admittance Matrices of Networks

The admittance matrix, a fundamental network analysis tool, relates current injections at a bus (node) to the bus (node) voltages. Kirchhoff's current law (KCL) requires that each of the current injections be equal to the sum of the currents flowing out of the node and into the lines connecting the node to other buses, or to the ground.

109

(13) ~~13~~

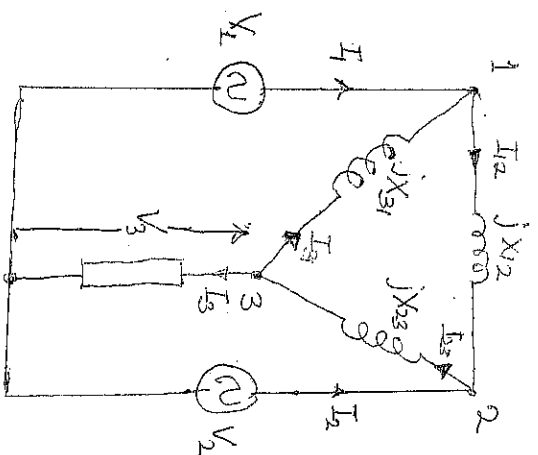


Fig. 19 Three-node network for power-flow analysis

Consider the network given in Fig. 1-9. With the voltages, currents, and line impedances as known, we can formulate the following equations for each node using Kirchhoff's laws: applying Kirchhoff's current law to nodes 1 to 3 yields:

Node 1:  $I_1 - I_{12} + I_{31} = 0 \dots (1)$

Node 2:  $I_2 - I_{23} + I_{12} = 0 \dots (2)$

Node 3:  $-I_3 + I_{23} - I_{31} = 0 \dots (3)$

Note that:  $I_{12} = \frac{(V_1 - V_2)}{jX_{12}}$

$I_{23} = \frac{(V_2 - V_3)}{jX_{23}}$ , and

$I_{31} = \frac{(V_3 - V_1)}{jX_{31}}$

Substituting for  $I_{12}$ ,  $I_{23}$  and  $I_{31}$  into equations (1), (2) and (3) gives:

Node 1:  $I_1 - \frac{(V_1 - V_2)}{jX_{12}} + \frac{(V_3 - V_1)}{jX_{31}} = 0 \dots (4)$

Node 2:  $I_2 - \frac{(V_2 - V_3)}{jX_{23}} + \frac{(V_1 - V_2)}{jX_{12}} = 0 \dots (5)$

Node 3:  $-I_3 + \frac{(V_2 - V_3)}{jX_{23}} - \frac{(V_3 - V_1)}{jX_{31}} = 0 \dots (6)$

But  $\frac{1}{jX} = Y = \text{admittance}$ , hence  $\frac{1}{jX_{12}} = Y_{12}$ ,  $\frac{1}{jX_{31}} = Y_{31}$ ,

and  $\frac{1}{K_{23}} = y_{23}$ .

Substituting  $y_{12}$ ,  $y_{31}$  and  $y_{23}$  into equations (4), (5) and (6) gives,

Node 1:  $I_1 - y_{12}(V_1 - V_2) + y_{31}(V_3 - V_1) = 0 \dots \dots (7)$

Node 2:  $I_2 - y_{23}(V_2 - V_3) + y_{12}(V_1 - V_2) = 0 \dots \dots (8)$

Node 3:  $-I_3 + y_{23}(V_2 - V_3) - y_{31}(V_3 - V_1) = 0 \dots \dots (9)$

Rewriting equations (7), (8) and (9) gives;

Node 1:  $I_1 = y_{12}(V_1 - V_2) - y_{31}(V_3 - V_1) \dots \dots (10)$

Node 2:  $I_2 = y_{23}(V_2 - V_3) - y_{12}(V_1 - V_2) \dots \dots (11)$

Node 3:  $-I_3 = -y_{23}(V_2 - V_3) + y_{31}(V_3 - V_1) \dots \dots (12)$

Expanding equations (10), (11) and (12) gives

$$I_1 = y_{12}V_1 - y_{12}V_2 - y_{31}V_3 + y_{31}V_1$$

$$I_2 = y_{23}V_2 - y_{23}V_3 - y_{12}V_1 + y_{12}V_2$$

$$-I_3 = -y_{23}V_2 + y_{23}V_3 + y_{31}V_3 - y_{31}V_1$$

Grouping like terms of the above equations together;

$$I_1 = (y_{12} + y_{31})V_1 - y_{12}V_2 - y_{31}V_3 \dots \dots (13)$$

$$I_2 = -y_{12}V_1 + (y_{12} + y_{23})V_2 - y_{23}V_3 \dots \dots (14)$$

$$-I_3 = -y_{31}V_1 - y_{23}V_2 + (y_{23} + y_{31})V_3 \dots \dots (15)$$

Equations (13), (14) and (15) can be put into matrix form as follows;

$$\begin{bmatrix} I_1 \\ I_2 \\ -I_3 \end{bmatrix} = \begin{bmatrix} (y_{12} + y_{31}) & -y_{12} & -y_{31} \\ -y_{12} & (y_{12} + y_{23}) & -y_{23} \\ -y_{31} & -y_{23} & (y_{23} + y_{31}) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

(15) (13)



It can be seen that the diagonal elements are the sum of the admittances connected to the node (Known as self-admittance) whilst the off-diagonal elements are the admittances connected between the relevant nodes, and are negative. Consequently, if the currents on the left-hand side are known, the voltages can be calculated by inverting the  $Y$  (nodal admittance) matrix. That is;

$$[Y][V] = [I], \text{ then } [V] = [Y]^{-1}[I] = \frac{[I]}{[Y]}$$

1.9.1

### Exercise -1

Formulate the nodal admittance matrix of the interconnected ~~buses~~ in Fig. 1.9.1

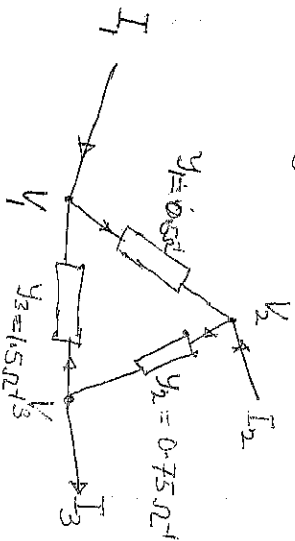


Fig. 1.9.1 Three busbar system load flows

### Solution

The nodal admittance matrix of the above network can be written by inspection as follows;

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} (Y_1 + Y_3) & -Y_1 & -Y_3 \\ -Y_1 & (Y_1 + Y_2) & -Y_2 \\ -Y_3 & -Y_2 & (Y_2 + Y_3) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

Replacing  $V_1, V_2$  and  $V_3$  in the matrix with their actual values gives;

$$\begin{bmatrix} I_1 \\ I_2 \\ -I_3 \end{bmatrix} = \begin{bmatrix} (0.5+1.5) & -0.5 & -1.5 \\ -0.5 & (0.5+0.75) & -0.75 \\ -1.5 & -0.75 & (0.75+1.5) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

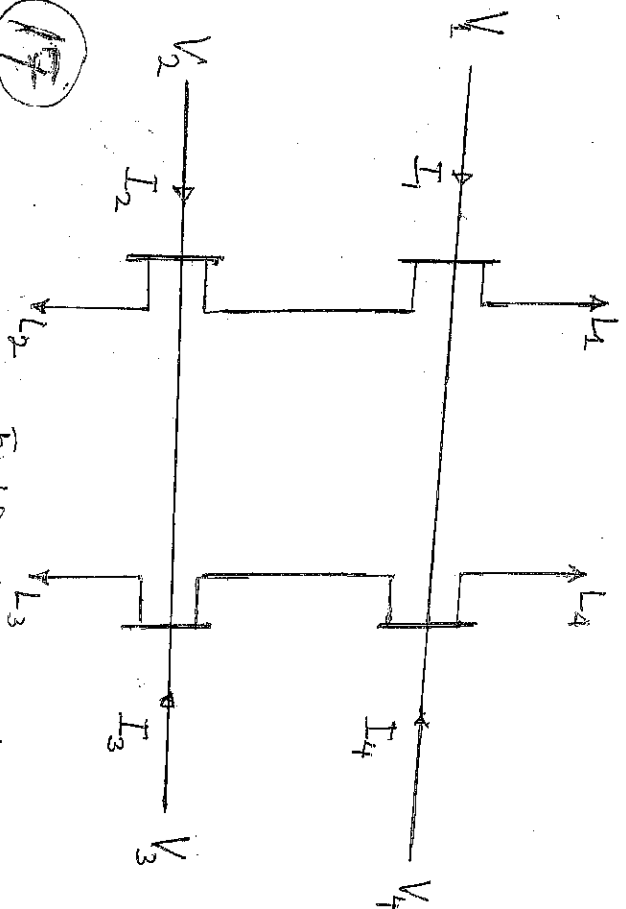
Hence, the final admittance matrix as shown below;

$$\begin{bmatrix} I_1 \\ I_2 \\ -I_3 \end{bmatrix} = \begin{bmatrix} 2 & -0.5 & -1.5 \\ -0.5 & 1.25 & -0.75 \\ -1.5 & -0.75 & 2.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

1.9.2

### Exercise - 3

- Given the interconnected power network in Fig. 1.9.2 where  $V_1, V_2, V_3, V_4, I_1, I_2, I_3$  and  $I_4$  are the injected voltages and currents at the buses while  $L_1, L_2, L_3$  and  $L_4$  are loads.
- (a) Draw the equivalent circuit of the network.
- (b) Form the equivalent circuit to formulate the nodal admittance matrix of the network.

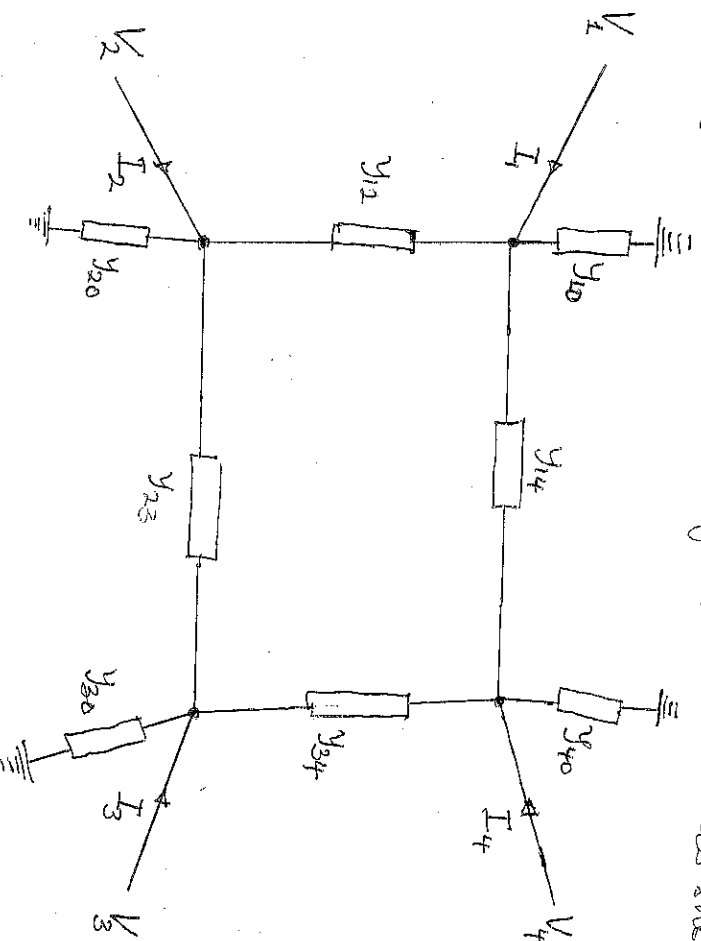


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Fig. 1.9.2 Interconnected power network

## Solution

The equivalent circuit of Fig. 1.9.2 is as shown below.



The nodal admittance matrix of the above network is as written hereunder.

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} (Y_{10} + Y_{12} + Y_{14}) & -Y_{12} & 0 & -Y_{14} \\ -Y_{12} & (Y_{20} + Y_{12} + Y_{23}) & -Y_{23} & 0 \\ 0 & -Y_{23} & (Y_{30} + Y_{23} + Y_{34}) & -Y_{34} \\ -Y_{14} & 0 & -Y_{34} & (Y_{40} + Y_{14} + Y_{34}) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

9.3

## Exercise - 1

Given the interconnected power network in Fig. 1.9.3 where  $V_1, V_2, V_3, V_4, I_1, I_2, I_3$  and  $I_4$  are the injected voltages and currents at the buses while  $L_1, L_2, L_3$  and  $L_4$  are loads.

- Draw the equivalent circuit of the network,
- From the equivalent circuit formulate the nodal admittance matrix of the network.

(18)

7

# Exercise 5

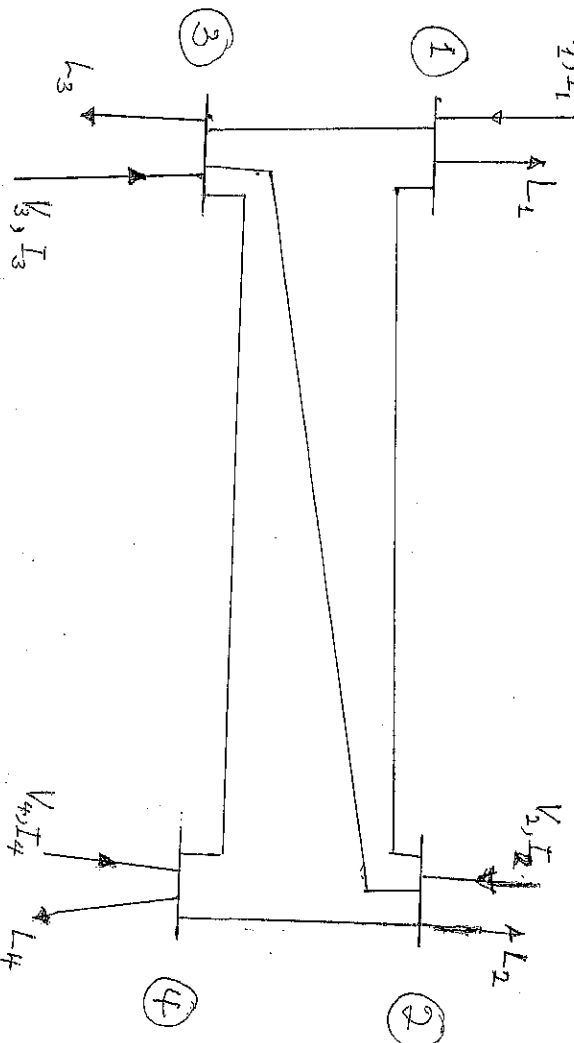


Fig. 1.9.3 Interconnected power system

The equivalent circuit of Fig. 1.9.3 is shown below.

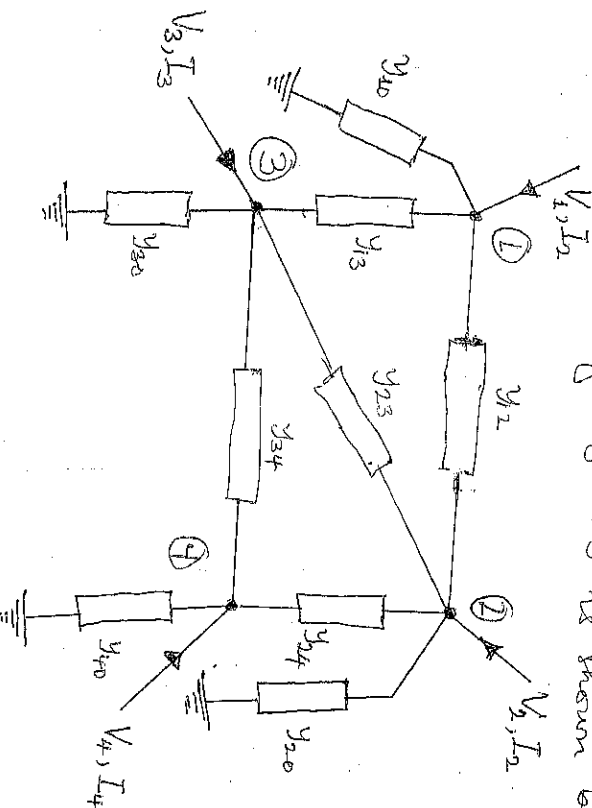


Fig. 1.9.4 Equivalent circuit of the interconnected power system in Fig. 1.9.3 and the nodal admittance matrix is as shown below.

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} (y_{10} + y_{12} + y_{13}) & -y_{12} & -y_{13} & 0 \\ -y_{12} & (y_{20} + y_{12} + y_{23} + y_{24}) & -y_{23} & -y_{24} \\ 0 & -y_{23} & (y_{30} + y_{13} + y_{23} + y_{34}) & -y_{34} \\ 0 & -y_{24} & -y_{34} & (y_{40} + y_{24} + y_{34}) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} \quad (19)$$

$$\text{Let } y_{11} = y_{10} + y_{12} + y_{13} + y_{14}$$

$$y_{22} = y_{20} + y_{21} + y_{22} + y_{23}$$

$$y_{33} = y_{30} + y_{31} + y_{32} + y_{33}$$

$$y_{44} = y_{40} + y_{41} + y_{42} + y_{43}$$

$$y_{23} = y_{32} = -y_{23}$$

$$y_{24} = y_{42} = -y_{24}$$

$$y_{34} = y_{43} = -y_{34}$$

Admittance matrix is written in terms of self-bus admittance  $y_{ii}$  and mutual bus admittance of  $y_{ik}$  as follows,

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} & y_{13} & y_{14} \\ y_{21} & y_{22} & y_{23} & y_{24} \\ y_{31} & y_{32} & y_{33} & y_{34} \\ y_{41} & y_{42} & y_{43} & y_{44} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

Hence, the bus admittance matrix is written as follows;

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & Y_{14} \\ Y_{21} & Y_{22} & Y_{23} & Y_{24} \\ Y_{31} & Y_{32} & Y_{33} & Y_{34} \\ Y_{41} & Y_{42} & Y_{43} & Y_{44} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}$$

In compact form  $I_{bus} = Y_{bus} V_{bus}$ .

#### 10. Need for Power Flow Studies:

The calculation in obtaining the steady-state condition of powers and voltages at various buses in power system network is known as load flow or power flow studies. Power flow studies are of the utmost importance because of the following reasons;

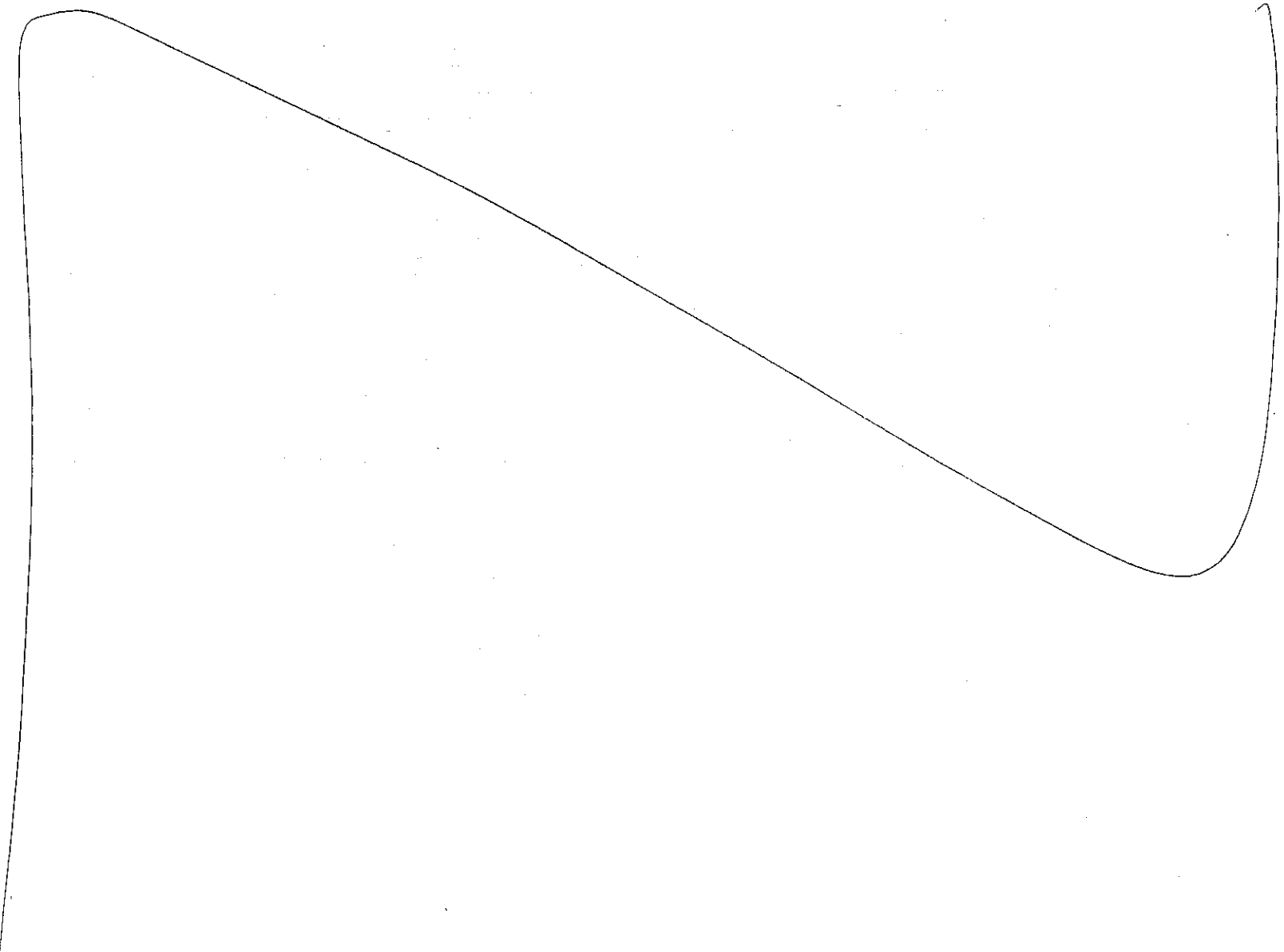
1. The studies frequently provide the starting condition for power system analysis such as transient stability, fault analysis and contingency analysis.
2. It is also used intensively in the planning of a new power system network or expansion of existing power system network.

Nowadays, load flow analysis is carried out almost extensively by digital computers and equations defining the problem are solved by special numerical techniques which have been developed to suit the special structure of the problem.

#### 10.1 Statement of Load Flow Problem:

For a given power network with known complex power loads and power set of specifications or restrictions on

power generations and voltages, solve for any unknown bus voltages and unspecified generation and finally for the complex power flow in the network components.



## Power Flow or Load Flow Study:

The ~~is the~~ study of various methods of solution to power system network is referred to as load flow study. The solution provides the bus:

- (a) magnitude of voltage ( $V$ ), (b) phase angle of voltage ( $\delta$ )
- (c) real power ( $P$ ), and (d) reactive power ( $Q$ ).

The load flow solution also gives the initial condition of the system when the transient behaviour of the system is to be studied.

### Need for Power Flow Study:

The load flow study of a power system is essential to decide the best operation of existing system and for planning the future expansion of the system. It is also essential for designing a power system.

### Statement of Power Flow Problem:

Each bus in a power system is associated with four quantities and they are real power ( $P$ ), reactive power ( $Q$ ), magnitude of voltage ( $V$ ) and phase angle of voltage ( $\delta$ ).

Work involved ( $n$ ) to be performed by a load flow study:

- (i) Representation of the system by a single line diagram;
- (ii) Determining the impedance (admittance) diagram using the information in single line diagram
- (iii) Formulation of network equations/Model Admittance matrix
- (iv) Solution of network equations

The load flow equations are non-linear algebraic equations and so explicit solutions are not possible. The solution of non-linear equations can be obtained only by iterative numerical methods.

### Bus or Node:

The meeting point of various components in a power system is called a "bus". The bus is a conductor made of copper or aluminium having negligible resistance. At some of the buses power is being injected into the network, whereas at other buses it is being tapped by the ~~load~~ system loads.



## Bus Classifications:

| Bus Classification                      | Specified Variable or Known Quantity | Unknown Variable or Quantity to be Determined |
|-----------------------------------------|--------------------------------------|-----------------------------------------------|
| Black or Swing or Reference bus         | $V, \delta$                          | $P, Q$                                        |
| Voltage-controlled, Generator or PV bus | $P, V$                               | $Q, \delta$                                   |
| Load or PQ bus                          | $P, Q$                               | $V, \delta$                                   |

## Load or PQ Nodes:

For  $P, Q$  nodes, the active and reactive powers ( $P, Q$ ) are specified as known parameters, while complex voltage magnitude,  $|V|$  and phase angle,  $\delta$  are to be solved for. Usually, substation nodes are taken as  $P, Q$  nodes.  $P, Q$  are the control variables while  $|V|, \delta$  are the dependent variables.

## PV Nodes:

For  $PV$  nodes, active power  $P$  and voltage magnitude,  $V$  are specified as known parameters, while reactive power,  $Q$  and voltage phase angle,  $\delta$  are to be determined.  $PV$  are the control variables while  $(Q, \delta)$  are the dependent variables. Generally speaking, the buses of power plants can be taken as  $PV$  nodes.

## Slack Node:

The slack node is specified by a voltage, constant in magnitude and phase angle. Therefore  $(V, \delta)$  node is the slack node.  $(V, \delta)$  are the control variables while active and reactive powers ( $P, Q$ ) are the dependent variables to be solved for. The effective generator at this node supplies the losses to the network.

## Power Flow Equations:

Power flow equation is as given below with generic Bus  $i$ :

$$P_i - jQ_i = \hat{V}^* \sum_{j=1}^n y_{ij} \hat{V}_j \quad i = 1, 2, 3, 4, \dots, n \quad (14)$$

The equation (18) represents a non-linear power flow equation.

Power flow equation (18) can be resolved into the real and reactive parts as follows;

$$P_i = \text{Real} \left[ \hat{V}_i^* \sum_{j=1}^n y_{ij} \hat{V}_j \right] \quad i = 1, 2, 3, \dots, n \quad (19)$$

$$Q_i = -\text{Imag} \left[ \hat{V}_i^* \sum_{j=1}^n y_{ij} \hat{V}_j \right] \quad i = 1, 2, 3, \dots, n \quad (20)$$

## General Form of Load Flow Equations:

The complex node voltage has two forms of representation depending on how the node voltages are represented. The forms are;

1. the polar form,
2. the rectangular form.

The node complex voltages are represented as follows,

$$\hat{V}_i = V_i \angle \theta_i$$

(21)

Equation (21) is resolved as;

$$\hat{V}_i = V_i (\cos \theta_i + j \sin \theta_i)$$

(22)

or

$$\hat{V}_i = e_i + j f_i$$

(23)

where  $e_i = V_i \cos \theta_i$  and  $f_i = V_i \sin \theta_i$

(25)

(24)

## Load Flow Equation in Polar Coordinates:

In polar ~~coord~~ coordinates, the real and reactive power injections are given by, where several nodes are involved;

$$P_i = V_i \sum_{j=1}^n V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij})$$

$$Q_i = V_i \sum_{j=1}^n V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \quad \left. \begin{matrix} i=1, 2, 3, \dots, n \\ (25) \end{matrix} \right\}$$

where  $G_{ij}$  is the conductance between buses  $i$  and  $j$ , and  $B_{ij}$  is the susceptance between buses  $i$  and  $j$ . Also  $\theta_{ij} = \theta_i - \theta_j$  which is the phase angle difference between bus voltages  $V_i$  and  $V_j$ , and  $n$  is the total number of buses.

## Load Flow Equation in Rectangular Coordinates:

In rectangular coordinates, the real and reactive power injections are given by,

$$P_i = \sum_{j=1}^n \left[ E_i (E_j G_{ij} - f_i B_{ij}) + f_i (f_j G_{ij} + E_j B_{ij}) \right]$$

$$Q_i = \sum_{j=1}^n \left[ f_i (E_j G_{ij} - f_j B_{ij}) - E_i (f_j G_{ij} + E_j B_{ij}) \right] \quad \left. \begin{matrix} i=1, 2, 3, \dots, n \\ (26) \end{matrix} \right\}$$

where  $G_{ij}$  and  $B_{ij}$  are as defined ~~not~~ above.

Note: Equations (25) and (26) are simultaneous non-linear equations of node power flows.

## Solution Techniques of Load Flow

The equations of power in load flow are non-linear simultaneous equations which cannot be solved with usual method but with iterative method. It entails repeatedly solving the equations until a convergent occurs which results into a solution.

The techniques are:

- (a) Gauss iterative method,
- (b) Gauss-Seidel iterative method,
- (c) Newton-Raphson iterative method (also referred to as hand-take method)

(d) Improved Newton-Raphson method.

(e) Fast decoupled method.

Many modifications and variations to these methods are also available. Most of these variations are introduced in terms of numerical techniques to either speed up the computation or reduce the required computer resources.

## Application of Digital Computers to Load Flow Studies

In load flow studies there are several mathematical calculations which consist of many steps depending on the size of the system. This process is difficult and takes a lot of times to perform by hand or manually, hence, the need of a fast method which the digital computer does.

Using digital computers to calculate load flow started from the middle of the 1950s. Since then, a variety of computer methods has been used in load flow calculation. The development of these methods is mainly led by the basic requirements of load flow calculation, which is summarized as follows,

1. The convergence properties of the method,
2. The computing efficiency and memory requirements of the method,
3. The convenience and flexibility of the implementation of the method.

As mentioned earlier, load flow problem is the task of solving a system of nonlinear algebraic equations which usually involves iteration process. Thus, reliable convergence becomes the prime criterion for the selection of load flow calculation method.

Newton-Raphson method is commonly used for load flow calculation because of its favorable convergence. Since the 1970s, the load flow calculating methods have continued to develop in various ways. Among the most successful is the "fast decoupled method" or also called P-Q decoupled method, this method is much simpler, and more efficient, and therefore more popular in many applications.

Let us,

$$I_R + I_Y + I_B = I_{R1} (1 + a^2 + a) + I_{R2} (1 + a + a^2) + I_{R0} (1 + 1 + 1)$$

$$\text{But } 1 + a + a^2 = 0, \text{ Hence,}$$

$$I_R + I_Y + I_B = I_{R1}(0) + I_{R2}(0) + 3I_{R0}, \text{ therefore,}$$

$$I_R + I_Y + I_B = 3I_{R0}, \text{ hence,}$$

$$I_{R0} = \frac{1}{3} (I_R + I_Y + I_B) \quad \text{--- (14)}$$

### (c) Positive Sequence Current

Multiplying expression (3) by "a" and expression (3) by  $a^2$  and then adding the results to expression (1) as follows;

$$\text{Expression (2) multiplied by "a" gives,}$$

$$(I_Y = a^2 I_{R1} + a I_{R2} + I_{R0}) \times a \text{ gives,}$$

$$a I_Y = a^3 I_{R1} + a^2 I_{R2} + a I_{R0}$$

$$\text{Expression (3) multiplied by } a^2, \text{ gives,} \quad \text{--- (15)}$$

$$(I_B = a I_{R1} + a^2 I_{R2} + I_{R0}) \times a^2 \text{ gives,}$$

$$a^2 I_B = a^3 I_{R1} + a^4 I_{R2} + a^2 I_{R0}$$

$$\text{Adding expressions (1), (15) and (16) together gives} \quad \text{--- (16)}$$

$$I_R + a I_Y + a^2 I_B = I_{R1} + I_{R2} + I_{R0} + a^3 I_{R1} + a^2 I_{R2} + a I_{R0} + a^2 I_{R1} + a^4 I_{R2} + a^2 I_{R0}$$

Collecting like terms on the right hand side together gives,

$$I_{R1} + a I_Y + a^2 I_B = (I_{R1} + a^3 I_{R1} + a^2 I_{R1}) + (I_{R2} + a^2 I_{R2} + a^4 I_{R2}) + (I_{R0} + a I_{R0} + a^2 I_{R0})$$

$$= I_{R1} (1 + a^3 + a^2) + I_{R2} (1 + a^2 + a^4) + I_{R0} (1 + a + a^2)$$

$$= I_{R1} (1 + 1 + 1) + I_{R2}(0) + I_{R0}(0)$$

$$= 3 I_{R1}$$

$$\therefore 3 I_{R1} = I_{R0} + a I_Y + a^2 I_B$$

$$I_{R1} = \frac{1}{3} (I_R + a I_Y + a^2 I_B)$$

(17)

(7)

### (c) Negative Sequence Current

Multiplying expressions (2) by  $a^{23}$  and expression (3) by  $a$  and then adding to expression (1) as follows;

$$(I_Y = a^2 I_{A1} + a I_{A2} + I_{A0}) \times a^3 = (a^5 I_Y = a^4 I_{A1} + a^3 I_{A2} + a^3 I_{A0})$$

Similarly,

$$(I_A = a I_{A1} + a^2 I_{A2} + I_{A0}) \times a = (a I_A = a^2 I_{A1} + a^3 I_{A2} + a I_{A0})$$

Adding the expression (1) + (a<sup>23</sup>I<sub>Y</sub> = a<sup>4</sup>I<sub>A1</sub> + a<sup>3</sup>I<sub>A2</sub> + a<sup>3</sup>I<sub>A0</sub>) + (a I<sub>A</sub> = a<sup>2</sup>I<sub>A1</sub> + a<sup>3</sup>I<sub>A2</sub> + a I<sub>A0</sub>) gives;

$I_A + a^2 I_Y + a I_A = I_{A1} + I_{A2} + I_{A0} + a^4 I_{A1} + a^3 I_{A2} + a^3 I_{A0} + a^2 I_{A1} + a^3 I_{A2} + a I_{A0}$

Collecting like terms on the R.H.S. together gives;

$$I_A + a^2 I_Y + a I_A = (I_{A1} + a^4 I_{A1} + a^2 I_{A1}) + (I_{A2} + a^3 I_{A2} + a^3 I_{A2}) + (I_{A0} + a^3 I_{A0} + a I_{A0})$$

$$= I_{A1}(0) + I_{A2}(3) + I_{A0}(0)$$

That is;

$$I_A + a^2 I_Y + a I_A = 3 I_{A2}, \text{ hence,}$$

$$I_{A2} = \frac{1}{3} (I_A + a^2 I_Y + a I_A) \quad \dots \dots \dots (8)$$

Equations (4), (7), and (8) are the expressions for the sequence currents or symmetrical components in terms of the values. The currents  $I_{R1}, I_{R2}$  and  $I_{R0}$  are the symmetrical components of the red phase (R-phase). Also,  $I_{Y1}, I_{Y2}$  and  $I_{Y0}$  are the symmetrical components of the yellow phase (Y-phase). Similarly,  $I_{B1}, I_{B2}$  and  $I_{B0}$  are the symmetrical components of the blue phase (B-phase). The analysis has been made considering currents; the method applied equally to voltages. Thus, the symmetrical voltage components of red-phase are;

$$E_{R0} = \frac{1}{3} (E_R + E_Y + E_B) \quad \dots \dots \dots (9)$$

$$E_{R1} = \frac{1}{3} (E_R + a E_Y + a^2 E_B) \quad \dots \dots \dots (10)$$

$$E_{R2} = \frac{1}{3} (E_R + a^2 E_Y + a E_B) \quad \dots \dots \dots (11)$$

(12)

## Facts About Sequence Components

- ① A balanced 3-phase system consists of positive sequence components only; the negative and zero sequence components are zero.
- ② The presence of negative or zero sequence components in a 3-phase system introduces unsymmetry and is indicative of an abnormal condition of the circuit in which these components are found.
- ③ The vector sum of the positive and negative sequence components of an unbalanced 3-phase system is zero. The resultant solely consists of three-zero sequence currents, namely;  
$$I_{e0} + I_{y0} + I_{a0}$$
- ④ In a 3-phase, 4-wire unbalanced system, the magnitude of zero sequence components is one-third of the current in the neutral conductor, that is;

Magnitude of Zero Sequence Current =  $\frac{1}{3}$  (Current in Neutral wire).

### Expression for Power in Symmetrical Components

The total power in symmetrical component is;

$3 \times$  (the sum of the individual sequence powers in any phase);

Hence, the complex,  $S$ , is given by;

$$S = 3 (I_0 E_0 + I_1 E_1 + I_2 E_2) \text{ Volt-Amperes}$$

where

$I_0$  and  $E_0$  are

zero sequence current and voltage;

$I_1$  and  $E_1$  are

positive " " " "

$I_2$  and  $E_2$  are

negative " " " "

in terms of real power or active power it is expressed as;

$$P = 3 (I_0 E_0 \cos \phi_0 + I_1 E_1 \cos \phi_1 + I_2 E_2 \cos \phi_2) \text{ Watts.} \quad \text{--- (13)}$$



### Example:

In a 3-phase, 4-wire system, the currents in R, Y and B lines are unbalanced conditions of loading are as follows;

$$I_R = 100 \angle 30^\circ \text{ A}; I_Y = 50 \angle 300^\circ \text{ A}; I_B = 30 \angle 180^\circ \text{ A}.$$

Calculate the positive, negative and zero sequence currents in the R-line and return current in the neutral wire.

Solution:

$$I_R = 100 \angle 30^\circ \text{ A}, I_Y = 50 \angle 300^\circ \text{ A}, I_B = 30 \angle 180^\circ \text{ A}.$$

Positive sequence current;

$$\begin{aligned} I_{R1} &= \frac{1}{3} (I_R + a I_Y + a^2 I_B) \\ &= \frac{1}{3} (100 \angle 30^\circ + 1 \angle 30^\circ \times 50 \angle 300^\circ + 1 \angle -120^\circ \times 30 \angle 180^\circ) \quad \text{Since } a = 1 \angle 120^\circ \\ &= \frac{1}{3} (100 \angle 30^\circ + 50 \angle 420^\circ + 30 \angle 60^\circ) \\ &= \frac{1}{3} [(86.6 + j50) + (45 + j43.3) + (15 + j25.98)] \\ &= \frac{1}{3} [146.6 + j117.28] = [48.9 + j39.76] \\ &= 54.98 \angle 43.3^\circ \text{ A} \end{aligned}$$

Negative sequence current;

$$\begin{aligned} I_{R2} &= \frac{1}{3} [I_R + a^2 I_Y + a I_B] \\ &= \frac{1}{3} [100 \angle 30^\circ + 1 \angle -120^\circ \times 50 \angle 300^\circ + 1 \angle 120^\circ \times 30 \angle 180^\circ] \\ &= \frac{1}{3} [100 \angle 30^\circ + 50 \angle 180^\circ + 30 \angle 200^\circ] \\ &= \frac{1}{3} [(86.6 + j50) + (-50 + j0) + (15 - j25.98)] \\ &= \frac{1}{3} [51.6 + j24.02] = [17.2 + j8] \end{aligned}$$

(17.2)

$$= 18.96 \angle 24.9^\circ \text{ A}$$

Zero Sequence Current;

$$\begin{aligned} I_{R0} &= \frac{1}{3} [I_R + I_Y + I_B] \\ &= \frac{1}{3} \left[ 100 \angle 30^\circ + 50 \angle 300^\circ + 30 \angle 180^\circ \right] \\ &= \frac{1}{3} \left[ (86.60 + j50) + (25 - j43.3) + (-30 + j0) \right] \\ &= \frac{1}{3} [81.6 + j6.7] = 27.2 + j2.23 \\ &= 27.29 \angle 4.65^\circ \text{ A} \end{aligned}$$

### Sequence Impedances of Power System Components

Each element of power system will offer different impedances to phase sequence components of current which may not be the same. For example, the impedance which any piece of equipment offers to positive sequence current will not necessarily be same as offered to negative sequence current or zero sequence current. Therefore, in unsymmetrical fault calculations, each piece of equipment will have three values of impedance, one corresponding to each sequence current  $I_{R1}, I_{R2}, I_{R0}$ .

1. positive sequence impedance,  $Z_1$ ;
2. negative sequence impedance,  $Z_2$ ;
3. Zero sequence impedance,  $Z_0$ ;

The sequence impedances of various components of power system (eg generator, transformers, transmission lines, etc) to positive, negative and zero sequence currents are of considerable importance in determining the fault sequence impedances of synchronous system.

### Sequence Impedances of Synchronous Generators

The positive, negative and zero sequence impedances of rotating machines are generally different. The positive sequence impedance of a synchronous generator is equal to the synchronous impedance of the machine. The negative sequence impedance is much less than the positive sequence impedance. The zero sequence impedance is a variable item and if its value is not given, it may be assumed to equal the positive

Sequence impedance: In summary,

- (1) Positive sequence impedance = synchronous impedance
- (2) Negative sequence impedance  $\neq$  Positive sequence impedance
- (3) Zero sequence impedance (if not given) = positive sequence impedance

Sequence Impedances of Transformers

Since transformers have the same impedance with reversed phase relation, their positive and negative sequence impedances are equal, thus being equal to the impedance of the transformer. However, the zero sequence impedance depends upon earth connection. If there is a through circuit for earth current, zero sequence impedance will be equal to positive sequence impedance. Otherwise it will be infinite. In summary,

- (1) Positive sequence impedance = Negative sequence impedance = Transformer impedance
  - (2) Zero sequence impedance = positive sequence impedance if there is earth circuit connection.
  - (3) Zero sequence impedance = infinite if there is no earth circuit connection.
- Sequence Impedance of Transmission Line

The positive sequence and negative sequence impedances of a line are the same; this value being equal to the normal impedance of the line. Zero sequence impedance is usually much greater than the positive and negative impedances. In summary;

- (1) Positive sequence impedance = Negative sequence impedance = Line impedance.
- (2) Zero sequence impedance  $\gg$  positive and negative sequence impedances.

Estimation of Fault Current With Sequence Impedances

In a 3-phase system, let the real phase be shorted to earth. Let  $E_1$ ,  $Z_1$ , and  $Z_0$  be positive, negative and zero sequence impedances respectively. Similarly, let  $I_1$ ,  $I_2$  and  $I_0$  be positive, negative and zero sequence currents.

The sequence currents will maintain voltage drops as their sequence impedances follow:

$$E_R = I_1 Z_1 + I_2 Z_2 + I_0 Z_0 + V_R$$

But  $V_R = 0$  since red phase is shorted to ground,

$$I_1 = I_2 = I_0 = \frac{1}{3} I_R, \text{ then } I_0 = \frac{1}{3} I_R \quad (14)$$

$$E_R = I_0 Z_1 + I_0 Z_2 + I_0 Z_0, \text{ then} \quad (15)$$

$$E_R = I_0 (Z_1 + Z_2 + Z_0)$$

$$\text{or } I_0 = \frac{E_R}{(Z_1 + Z_2 + Z_0)} \quad (16)$$

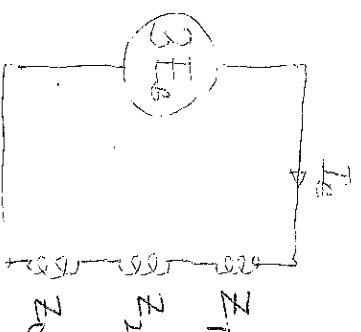
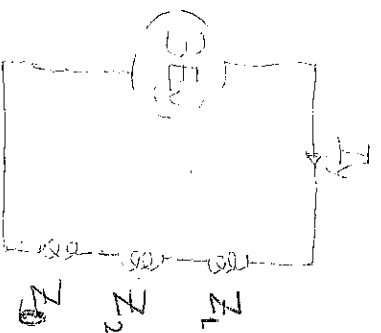
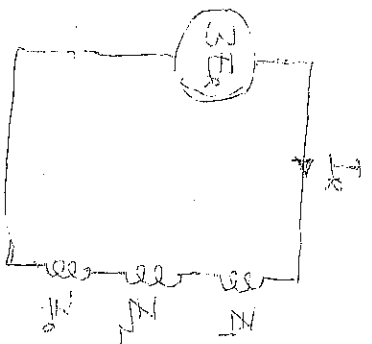
(17)

From eqn (15), we noted that  $I_0 = \frac{1}{3} \cdot I_R$ , substituting for  $I_0$  into eqn (17), gives;

$$\frac{1}{3} \cdot I_R = \frac{E_R}{(Z_1 + Z_2 + Z_0)} \quad \text{hence fault current is}$$

$$I_R(I_R^F) = \frac{3E_R}{(Z_1 + Z_2 + Z_0)} \quad \text{--- (18)}$$

From equation (18), the equivalent circuit from which fault current may be calculated from ohm's law, where, yellow-phase and blue-phase are as shown below;



It is clear that fault current calculation equivalent circuit for a single phase to ground fault may be calculated by connecting the sequence impedances in series across imaginary generator of voltage  $3E$ . However, if the fault impedance is assumed to be zero. for fault current is modified as

$$I_R(I_{RF}) = \frac{3E_R}{(Z_1 + Z_2 + Z_0 + 3Z_0)} \quad \text{--- (19)}$$

$$I_Y(I_{YF}) = \frac{3E_Y}{(Z_1 + Z_2 + Z_0)} \quad \text{--- (20)}$$

$$I_0(I_{0F}) = \frac{3E_0}{(Z_1 + Z_2 + Z_0)} \quad \text{--- (21)}$$

(52)

## Example

A 3-phase, 10MVA, 11KV generator with a solidly earthed neutral point supplies a feeder. The relevant impedances of the generator and feeder in Ohms are as follows:

|                              | Generator | Feeder |
|------------------------------|-----------|--------|
| Positive sequence impedance: | $j1.2$    | $j1.0$ |
| Negative sequence impedance: | $j0.9$    | $j1.0$ |
| Zero sequence impedance:     | $j0.4$    | $j3.0$ |

If a fault from one phase to earth occurs on the far end of the feeder of red phase, calculate the magnitude of the fault current and line to neutral voltage at the generator terminals.

Solution:

$$(MVA)_{base} = 10, (KV)_{base} = 11$$

Positive sequence impedance,  $Z_1 = j1.2 + j1.0 = j2.2 \Omega$

Negative sequence impedance,  $Z_2 = j0.9 + j1.0 = j1.9 \Omega$

Zero sequence impedance,  $Z_0 = j0.4 + j3.0 = j3.4 \Omega$

The phase e.m.f. of R-phase,  $E_R =$

$$\frac{11 \times 10^3}{\sqrt{3}} = 6351$$

Since the fault is single line to ground fault, we have;

$$I_1 = I_2 = I_0 = \frac{E_R}{(Z_1 + Z_2 + Z_0)} = \frac{6351}{(j2.2 + j1.9 + j3.4)}$$

$$\text{i.e. } I_0 = \frac{6351}{j7.5} = \frac{j6351}{j7.5}$$

$$I_0 = \frac{j6351}{-7.5} = -j846.8$$

But fault current  $I_R = (I_f)$

$$= 3I_0 = 3 \times -j846.8 = -j2540.4$$

The magnitude of fault current,  $I_{kf} = \sqrt{(-j2540.4)^2}$

$$= \underline{2540.4A}$$

Line-to-neutral voltage of R-phase,  $V_R$

$$V_R = E_R - I_0(Z_1 + Z_2 + Z_0)$$

$$= 6351 - (-j846.8)(j1.2 + j0.9 + j0.4)$$

$$= 6351 + (j846.8)(j2.5)$$

$$= 6351 - 2117$$

$$= \underline{4234V} = \underline{4.234KV}$$

Sequence Networks

The analysis of an unsymmetrical fault by symmetrical components method can be conveniently done by drawing sequence networks. A sequence network of a particular sequence current in a given power system is the path for flow of that sequence current in the system. It is composed of impedances offered to that sequence current in the system. The sequence networks of a power system are;

1. Positive sequence network,
2. Negative sequence network, and
3. Zero sequence network.

2.24.1

Positive Sequence Network

The positive sequence network for a given power system shows all paths for the flow of positive sequence currents in the system. It is represented by one line diagram and is composed of impedances offered to the positive sequence currents. While drawing the positive sequence network of a given power system, the following points may be kept in mind:

1. Each generator in the system is represented by the generated voltage in series with reactance and resistance.
2. Current limiting impedances between the generator neutral and ground pass no positive sequence current and hence are not included in the positive sequence network.
3. All resistances and magnetising currents for each transformer are neglected.
4. For transmission lines, the shunt capacitance and resistances are generally neglected.
5. Motor loads are included in the network as generated e.m.f.s in series with appropriate reactance.

(70)

(22)

## Negative Sequence Network

The negative sequence network for a given power system shows all paths for the flow of negative sequence currents in the system. It is also represented by one-line diagram and is composed of the impedances offered to the negative sequence currents. The negative sequence network can be readily obtained from positive sequence network with the following modifications:

1. Omit the emfs of 3-phase generators and motors in the positive sequence network. It is because these devices have only positive sequence-generated components.
2. Change, if necessary, the impedances that represent rotating machinery in the positive sequence network. It is because negative sequence impedance of rotating machinery is generally different from that of positive sequence impedance.
3. Current limiting impedances between generators ~~and~~ neutral and ground pass no negative sequence current and hence are not included in the negative sequence network.
4. For static devices such as transmission lines and transformers, the negative sequence impedances have the same value as the corresponding positive sequence impedances.

## Zero Sequence Network

The zero sequence network for a given power system shows all the paths for the flow of zero sequence currents. The zero sequence network of a system depends upon the nature of connections of the 3-phase

(71)

(22)

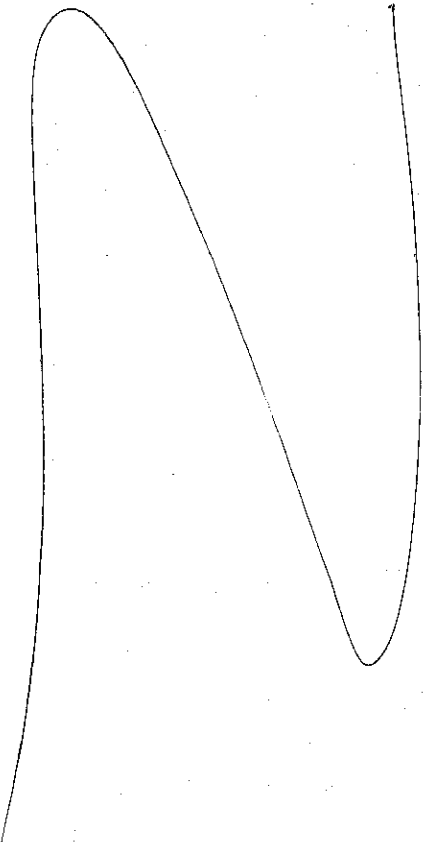
windings of the components in the system. The following points may be noted about zero sequence networks;

1. the zero sequence currents will flow only if there is a return path i.e. path from neutral to ground or to another neutral point in the circuit.
2. In the case of a system with no return path for zero sequence currents, these currents cannot exist.

#### 2.24.4 Reference Bus for Sequence Networks

While drawing the sequence networks, it is necessary to specify the reference potential with respect to which all sequence voltage drops are to be taken. For this purpose, the following points should be considered;

1. For positive or negative sequence networks, the neutral of the generator is taken as the reference bus. This is logical because positive or negative sequence components represent balanced sets and hence all the neutral points must be at the same potential for either positive or negative sequence currents.
2. For zero sequence network, the reference bus is the ground at the generator.



(72)

(24)



# Individual Sequence Networks for each Network Element

| SN  | Network Element | Positive | Negative | Zero |
|-----|-----------------|----------|----------|------|
| 1.  | Generator<br>   |          |          |      |
| 2.  | Transformer<br> |          |          |      |
| 3.  | Transformer<br> |          |          |      |
| 4.  | Transformer<br> |          |          |      |
| 5.  | Transformer<br> |          |          |      |
| 6.  | Transformer<br> |          |          |      |
| 7.  | Transformer<br> |          |          |      |
| 8.  | Transformer<br> |          |          |      |
| 9.  | Transformer<br> |          |          |      |
| 10. | Transformer<br> |          |          |      |

33

(2.5)

|    |                       |  |  |  |
|----|-----------------------|--|--|--|
| 11 | Transformer<br>       |  |  |  |
| 12 | Transmission Line<br> |  |  |  |
| 13 | Transmission Line<br> |  |  |  |
| 14 | Electric Motor<br>    |  |  |  |
| 15 | Electric Motor<br>    |  |  |  |

### Constructing Sequence Network

#### Example 1

Consider the power network shown on Fig. 2.25(a) Construct the sequence networks of the system.

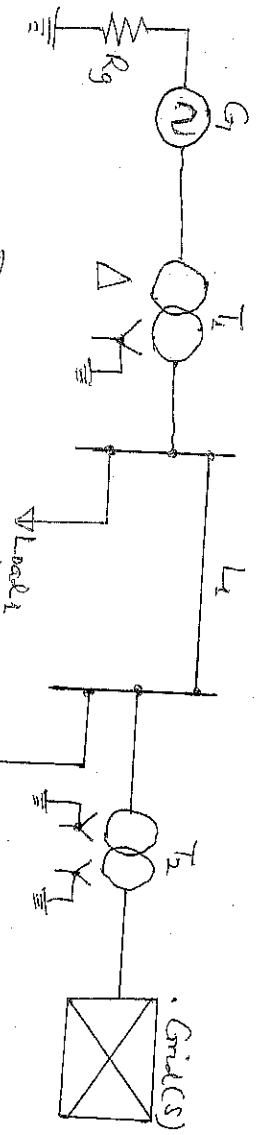
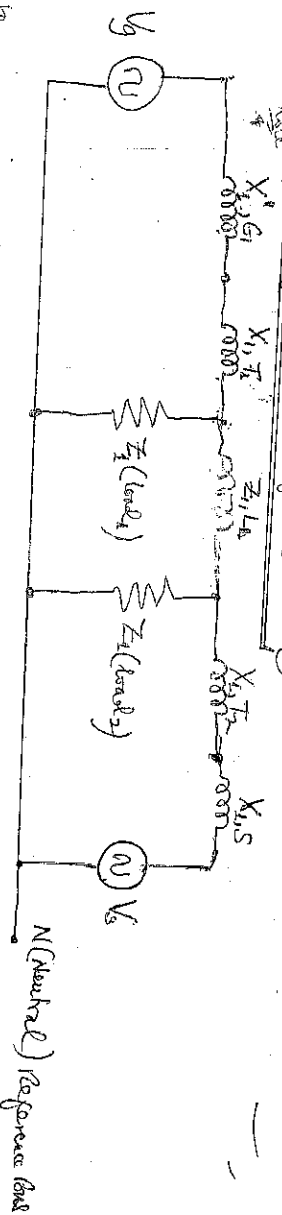


Fig. 2.25(a) Power System Network

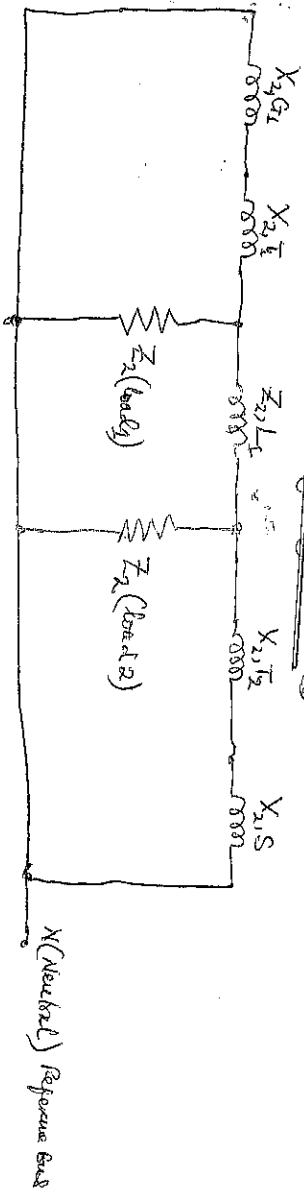
These individual networks of each element are then connected together to form the positive, negative and zero sequence networks for the overall system as follows:

#### Positive Sequence Network of Fig. 2.25(a)

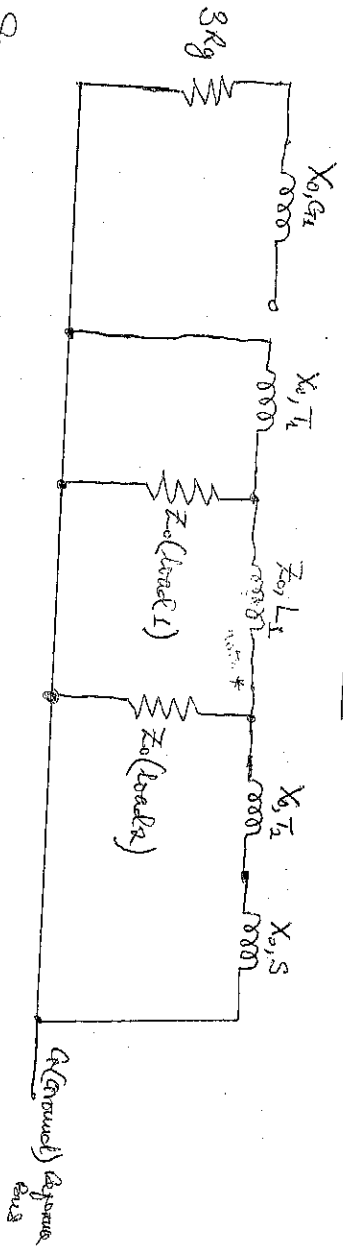


(26)

(b) Negative Sequence Network of Fig. 2.25(a)



(c) Zero Sequence Network of Fig. 2.25(a)



Example 2

Given the power system of Fig. 2.25(b) find the sequence networks.

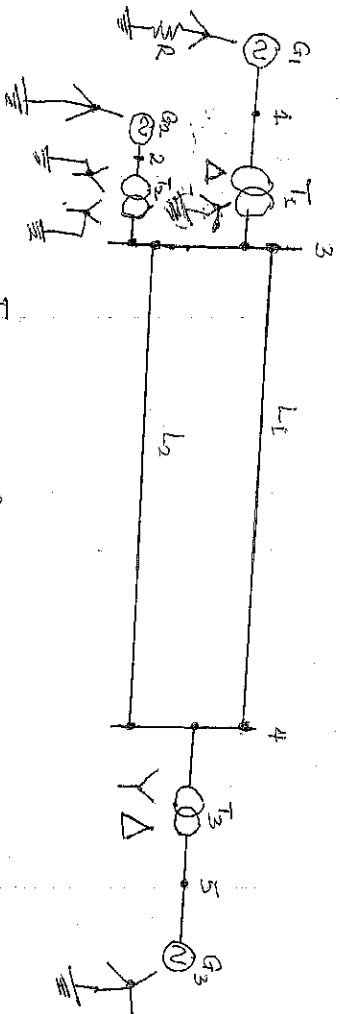


Fig. 2.25(b) Power System Network.

The sequence networks for the power network given in Fig. 2.25(b) are as shown below;

(a) Positive Sequence Network

